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Question 5 "Pseudocode"

Start

1 // Function Display menu( )
Printf " LIBRARY Management System"
Printf " All Choices"

2 // Registration
reg-mem (name[], *std_id, *mem_type)
Scanf("%[^\n]", name)
Scanf ("—", std-id, mem-type)
End function

3 // Display Books
dis-books (book_ids, book_titles, avlble_copies, total copies)
Printf ("ID | Title | Copies")
for (int i=0; i<4 ; i++)
{
printf("|%-4d |%-21d |%-4d| \n", book_ids,--)
} //End for

4 // Borrow Books
boorrow_books (bk_ids, bk_titles, avlbl_copies, ttl_cps brrd_book_ids , brrd_cant, mem_type , cant)
int book_id, days
Scanf ("%d %d", Bk_id, days)
if (bk_id < 101 || bk_id > 104)
{
Printf ("Enter Valid book Id")
} //End If
for (i=0, i<4, i++)
{

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```
if (mem_type ==1){
    if (bk_id == book_ids[i]{
        if(avlb_copies [i] >0) {
            if (days <=15)
            {
                Printf("you can take book")
                (*brrwd_ coost)++;
                brr-bk_id [count] = bk_ids[i];
                total_cpies --;
                avlbl_cpies [i] --
            } else {
                prtf("mem_lit 15 days")
            } //End if
        } Else {
            printf("No free book")
        }//End if
    }Else if (mem_type ==2) {
        if(bk_id == bk_ids [i]{
            if (avlb_copies [i] >0){
                if (days <=30){
                    Printf("you can take book")
                    (*brrwd_cowt)++.
                    brr_bk_id [count]=bk_ids[i]
                    tbl-cpies --
                }Else {
                printf("mem limit 30 days");
                }
            }Else{
            Printf("No copies available")
        }
```

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```
}Else {
    Printf ("Enter valid member type")
}// Return Books
float return_Book(bk_ids, avlbl_cpies, ttl_bks, brrd_bk_ids,
brrd_count, mem_type, total_fine)
    int bk_id, days, found=0, pos=0, allowed, rate;
    float fine=0, payable=0
    char code[20];
    Printf("Enter book id & issued days")
    Scanf("%d %d", &bk_id, &days)
    if (day<=0){
        printf("No must be +ve")
    }// Del if
    for (i=0; i<*brrd_count; i++){
        if (brrd_bkid_ids[i] == bk_id){
            found=1;
            Pos=1;
            break;
        }
    }//End for
    for (i=0; i<ttl_bks; i++){
        if (bk_IDs[i]==bk_id){
            avlbl_cpies[i]++
        }
    }// Del for
    if (mem_type==1){
        allowed=14
        rate=20
    }Else{
        allowed=30
```

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```
if (days > allowed) {
    fine = (days - allowed) * rate
    Printf("Fine is %f", fine)
    Printf("Enter membership Code")
    Scanf("%s", code)
    if (strcmp(code, "LIB2026") == 0) {
        payable = fine/2
    } else {
        payable = fine;
    } // End if
    Printf("Fine is %f", payable)
} Else {
    Printf("No fine")
}
*total_fine += fine
for (i = pos; i < *brrd_count - 1; i++) {
    brrd_bk_ids[i] = brrd_bk_ids[i+1];
}
(*brrd_count)--
return payable
} // End fuction
{ // display Member Record
display_memb_record(name, ID, mem_type, brrd_bk_Ids
brrd_count, toll fine, fineattenuer) {
    Printf("Name: %s", name)
    Printf("ID %d \n", StdID)
    if (memb_type == 1) {
        printf("Member type: Student")
    } Else {
        Printf("Faculty")
```

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```
if (brrd_cad ==0 ){
    printf("None")
}
for (i=0; i< brrd_caud ; i++) {
    printf ("%d", brrd_book_Ids [i]:
} //End for
Printf("Total fine: %f ", total fine):
Printf("Final fine %f ", fine after waiver)
```

7. Main function

```
int main () {
    char title [4][50] ={"Programing in C", "Data structures",
"Database Systems", "Computer Networks" };
    int total copies =0;
    int copies [4];
    Cps [0] = 5;
    Cps[1] = 3;
    Cps[2) = 0;
    Cps [3] = 4;
    InA Ids [4];
    for (i=0, i<4, i++){
        Ids [i] = 101+i;
        totalcopies += Copies [i];
} //End for
int borrowed_ids [3], borrowed_cout =0, Std_id, new_typ;
char Trans [100];
float total_fine =0, fine_after =0;
int choice, registered =0, Cout =0,
```

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```
do
{
  display menu( );
  Scanf("%d", choice);
  Switch (choice)
  {
  Case 1:
      reg-mem(name, std-id, mem_type);
      if (mem_type = 1 || mem_type ==2)
      {
          registered == 1;
      } Else {
          registered =0;
      }
      break;
  Case 2:
      display_books (Ids, title, copies, ttl_copies);
      break;
  Case 3;
      if (registerd ==0) {
          Printf("register first");
      } Else if (brrowed_card >=3) {
          Prtf("borrow max 3 books");
  Case 3: Else {
          Count= brrwed_card;
          borrow-books (Ids, title, copies ____ );
      }
      break;
```


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```
Case 4;
    if (registered ==0)
        printf( register first)
    Else if (borrowed_count == 0)
        Print (Not borrowed any book)
    Else
        fine_after_varies += returnBook (Ids, cpees, 4,)
    break;
Case 5:
    if (reg==0)
        prithf (register first)
    Else
        displayMemberRecord (names, std_id, mem_type
Case 6:
    Prat ("Existing")
    brak;
Default:
    Pralf (Invalid Choice)
}
} while (choice != 6);
```

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